



## M002 Week 7 — English Worksheet

**Topic:** Pistons and Mazes · **Course:** Maze Madness · **Time:** about 45 minutes

This week the maze uses **pistons**. A piston pushes or pulls a block when it gets a redstone signal. A **sticky piston** can pull a block back, opening a path. Your agent flips a lever to power a piston, then crosses the gap it opens.

### 1 · Predict 🧠

Read each set of steps. Before you imagine the agent doing it, write what you think will happen.

```
move forward
interact ahead
move forward
```

The agent walks up to a lever, flips it, and a sticky piston pulls a block back. What happens to the path? Can the agent move forward now?

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```
if gap ahead:
    interact ahead
otherwise:
    move forward
```

There is a gap in the floor with a piston bridge. What does the agent do at the gap? What does it do on solid ground?

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```
keep doing forever:
  if gap ahead:
    interact ahead
    move forward
  otherwise:
    move forward
```

**Describe in plain English what the agent is doing, step by step.**

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## 2 · Spot the Bug 🐛

Each block of code below was meant to do something, but it is broken. Read what the code is **supposed** to do, then rewrite it so it works. After that, explain why the original was wrong and why your fix works.

**Bug A** — The agent is supposed to **power the piston first**, then cross the gap it opens.

```
move forward
move forward
interact ahead
```

**Hint:** flip the lever *before* you reach the gap, not after.

**Write the fixed code:**

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**Why was it wrong? Why does your fix work?**

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**Bug B** — The agent should only **interact** when there is a **gap** ahead. Right now it tries to flip a lever on every step.

```
keep doing forever:
  interact ahead
  move forward
```

**Write the fixed code:**

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**Why was it wrong? Why does your fix work?**

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**Bug C** — A sticky piston holds the bridge **only while the lever is ON**. The agent must cross **before** turning the lever back off.

```
interact ahead  
move forward  
interact ahead  
move forward
```

**Hint:** the second `interact` switches the lever OFF and the bridge disappears. Cross first.

**Write the fixed code:**

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**Why was it wrong? Why does your fix work?**

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### 3 · Tell Me What You Built 📷

Now switch to your homework world. The maze has a **gap** (or a blocked path) and a **piston** that opens it when powered. Write code that:

1. moves the agent to the lever,
2. flips it ( **interact** ) so the piston opens the path,
3. crosses while the path is open,
4. reaches the goal.

When you finish, come back here. Send a photo or video of the agent reaching the end, then explain what you did. Use these sentence starters — write 4 to 6 sentences total.

First, I walked the agent to ...

I powered the piston by ...

Once the path opened, the agent ...

The hardest part was ...

To fix it, I ...

If I had more time, I would ...

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## 4 · Record Your Walkthrough

Now take a video on your phone (or a parent's phone) while the agent runs the maze. Talk through it like you are teaching someone who has never seen it. Try to use these words: **piston**, **redstone**, **signal**, **gap**, **bridge**.

1. Show the gap (or blocked path) and the piston in the maze.
2. Run your code and show the agent powering the piston and crossing.
3. Read your code out loud and say which line opens the path.
4. Show one bug you hit and how you fixed it.

**Write what you will say in your video.** Use the space below to plan it before you record — you can read from it while filming.

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**Submit** 

Send this worksheet + your walkthrough video to teacher on KakaoTalk.